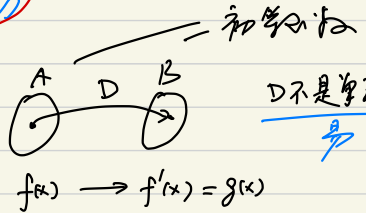


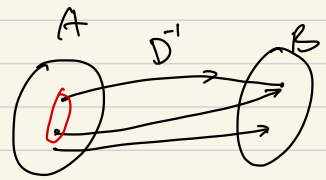
第二章 导数 与 微分

求导 是 γ 映射.
D



D 不是单射. 也不是满射.
易 难

初等函数的导数, 若不是初等函数.
例. e^{-x^2}



$\{ \underbrace{f_1, f_2, \dots, f_n, \dots}_{\text{合年式等所类}} \}$

记号

$\int f(x) dx = F(x) + C$ → 理解为一个集合.

$$\frac{dF(x)}{dx} = f(x)$$

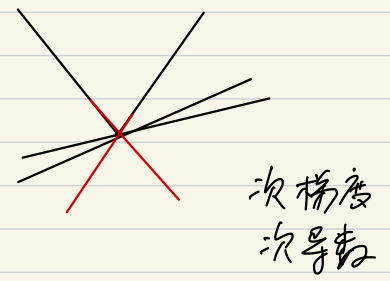
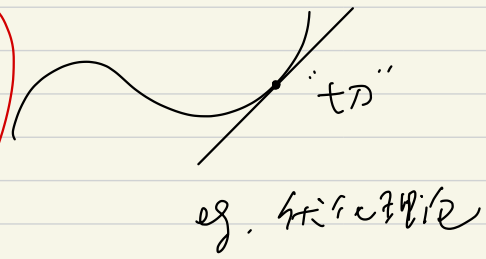
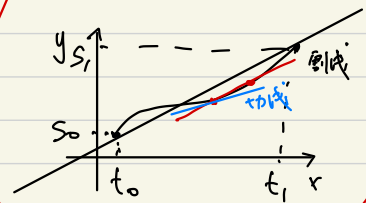
$$dF(x) = f(x) dx$$

$$\int dF(x) = \int f(x) dx$$

$$\begin{aligned} &F(x) - F(x_0) \\ &= \\ &F(x) + C \end{aligned}$$

§ 2.1 导数概念.

何可定义导数.



① 定义: $f(x)$ 在 x_0 处的导数.

切线方程: $y - f(x_0) = f'(x_0)(x - x_0)$

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

存在, 可导 $\rightarrow f'(x_0), \frac{df(x)}{dx} \Big|_{x=x_0}, \frac{df(x_0)}{dx}, y'$

不存在, 不可导 $\rightarrow$ $\left. \begin{array}{l} \text{左极限存在} \rightarrow \text{左导数} \\ \text{右极限存在} \rightarrow \text{右导数} \end{array} \right\}$ 单侧导数
左右极限都不存在

$$\parallel \frac{\Delta f}{\Delta x}, \frac{\Delta y}{\Delta x}$$

$$\Delta \rightarrow d \rightarrow \frac{df}{dx}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = A \not\Rightarrow \lim_{\Delta x \rightarrow 0} \Delta y = 0$$

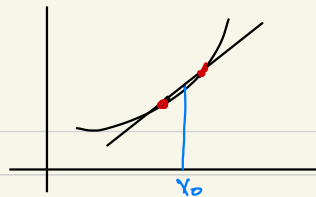
$\lim_{\Delta x \rightarrow 0} \Delta x = 0$

可导 $\nRightarrow$ 连续

$$\textcircled{+} \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x} = f'(x_0)$$

注: 不可作为导数的定义!

例 $f(x) = |x|$, $x_0 = 0$. 极限 $= 0$, 但 $f'(x_0)$ 不存在.



$\textcircled{+}$ $|x|$ 在 0 不可导 (但连续).

$\textcircled{+}$ 导函数.

$\textcircled{+}$ 存在处处连续但处处不可导的函数 (将来会学)

② 导数的计算

1. 基本初等函数

2. $+ - \times \div 0$. 运算

$$\begin{cases}
 1^\circ f(x) = C. & f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \rightarrow 0} 0 = 0 \\
 2^\circ f(x) = x^n. & f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{(x_0 + \Delta x)^n - x_0^n}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(\cancel{x_0^n} + n \cancel{x_0^{n-1}} \Delta x + \dots + \Delta x^n) - \cancel{x_0^n}}{\Delta x} = n x_0^{n-1} \\
 3^\circ f(x) = x^\mu & f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{(x_0 + \Delta x)^\mu - x_0^\mu}{\Delta x} = x_0^{\mu-1} \lim_{\Delta x \rightarrow 0} \frac{(1 + \frac{\Delta x}{x_0})^\mu - 1}{\frac{\Delta x}{x_0}} = \mu x_0^{\mu-1} \\
 & \quad \downarrow x \rightarrow e^{\mu \ln x} \\
 \text{eg. } f(x) = \sqrt{x}. & f'(x) = x^{-\frac{1}{2}} \\
 4^\circ f(x) = \sin x. & f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\sin(x_0 + \Delta x) - \sin(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\cos(x_0 + \frac{\Delta x}{2}) \sin(\frac{\Delta x}{2})}{\Delta x/2} = \cos(x_0) \\
 5^\circ f(x) = a^x. & f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{a^{x_0 + \Delta x} - a^{x_0}}{\Delta x} = a^{x_0} \lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = a^{x_0} \ln a \\
 6^\circ f(x) = \log_a x. & f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\log_a(x_0 + \Delta x) - \log_a(x_0)}{\Delta x} = \frac{1}{x_0} \lim_{\Delta x \rightarrow 0} \frac{\log_a(1 + \frac{\Delta x}{x_0})}{\frac{\Delta x}{x_0}} = \frac{1}{x_0} \frac{1}{\ln a} \\
 7^\circ \arcsin(x). & \left. \begin{array}{l} \\ \end{array} \right\} \text{用运算法则.} \\
 8^\circ \dots & \left. \begin{array}{l} \\ \end{array} \right\}
 \end{cases}$$

2022.10.14
Fri

§2.2. 函数的求导法则

$+$ $-$ $\times$ $\div$ 0 -1

① 四则运算 $+$ $-$ $\times$ $\div$

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

$$(f(x) \cdot g(x))' = \cancel{f'(x) g(x)} ?$$

$$\lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x)g(x+\Delta x) - f(x)g(x)}{\Delta x} = \underbrace{\frac{f(x+\Delta x)(g(x+\Delta x) - g(x))}{\Delta x}}_{f(x) \cdot g'(x)} + \underbrace{\frac{(f(x+\Delta x) - f(x))g(x)}{\Delta x}}_{f'(x) \cdot g(x)}$$

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\left(\frac{1}{g(x)}\right)' = ? \quad \cancel{\frac{1}{g'(x)}}$$

$$\left(\frac{1}{g(x)}\right)' = -\frac{g'(x)}{g^2(x)}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\frac{1}{g(x+\Delta x)} - \frac{1}{g(x)}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{g(x) - g(x+\Delta x)}{g(x+\Delta x)g(x)}}{\Delta x} = \frac{-g'(x)}{g^2(x)}$$

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

$$\frac{\cancel{g(x+\Delta x) - g(x)}}{\cancel{g(x+\Delta x)g(x)}} \cdot \frac{-1}{\Delta x} = \frac{-g'(x)}{g^2(x)}$$

② 复合. $f'(x), g'(x)$

链式求导

chain rule

$$(f \circ g(x))' = (f(g(x)))' = f'(g(x)) g'(x)$$

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{f(g(x+\Delta x)) - f(g(x))}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{f(\underbrace{g(x+\Delta x)}_{=u+\Delta u}) - f(\underbrace{g(x)}_{=u})}{\underbrace{g(x+\Delta x) - g(x)}_{\Delta u}} \cdot \frac{g(x+\Delta x) - g(x)}{\Delta x} \\ &= \lim_{\Delta u \rightarrow 0} \frac{f(u+\Delta u) - f(u)}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{g(x+\Delta x) - g(x)}{\Delta x} \\ &= f'(u) g'(x) = f'(g(x)) g'(x) \end{aligned}$$

$$(f_1 \circ f_2 \circ f_3 \circ \dots \circ f_n(x))' = f_1' \cdot f_2' \cdot \dots \cdot f_n'(x)$$

③ 反函数. $(f^{-1}(y))'$

$$\begin{aligned} y &= f(x) \\ x &= f^{-1}(y) \end{aligned}$$



条件: $f(x)$ 附近可导

$$\lim_{\Delta y \rightarrow 0} \frac{f^{-1}(y+\Delta y) - f^{-1}(y)}{\Delta y} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta y} = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta y / \Delta x} = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))}$$

1.1. (f)

$$\left\{ \begin{array}{l} (f \pm g)' = f' \pm g' \\ (fg)' = f'g + fg' \quad , \quad \frac{d(fg)}{dx} = \frac{df}{dx}g + f \frac{dg}{dx} \\ (1/g)' = -g'/g^2 \\ (f/g)' = \frac{f'g - fg'}{g^2} \\ (f \circ g)' = f' \cdot g' \quad \leftarrow \quad \frac{d(f \circ g)}{dx} = \frac{d(f \circ g)}{dg} \frac{dg}{dx} \\ (f^{-1})' = \frac{1}{f'} \quad \leftarrow \quad \frac{dx}{dy} = \frac{1}{dy/dx} \end{array} \right.$$

$$\sin'(x) = \cos(x) \quad , \quad (x^u)' = u x^{u-1} \quad , \quad (a^x)' = \ln a \cdot a^x$$

$$\text{Ex: } (\cos x)' = \left(\sin\left(x + \frac{\pi}{2}\right) \right)' = \sin'\left(x + \frac{\pi}{2}\right) \cdot 1 = \cos\left(x + \frac{\pi}{2}\right) = -\sin(x)$$

$$\begin{aligned} (\tan(x))' &= \left(\frac{\sin(x)}{\cos(x)} \right)' = \frac{\sin'(x) \cos(x) - \sin(x) \cos'(x)}{\cos^2 x} \\ &= \frac{\cos^2(x) + \sin^2(x)}{\cos^2 x} = \frac{1}{\cos^2(x)} = \sec^2(x) \end{aligned}$$

$$(\cot(x))' = \left(\frac{\cos(x)}{\sin(x)} \right)' = -\frac{1}{\sin^2(x)} = -\csc^2(x)$$

$$(\sec(x))' = \left(\frac{1}{\cos(x)} \right)' = -\frac{-\sin(x)}{\cos^2(x)} = \sec(x) \cdot \tan(x)$$

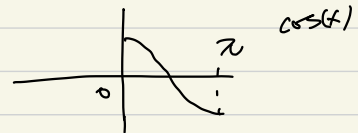
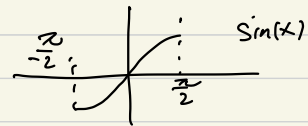
$$(\csc(x))' = \left(\frac{1}{\sin(x)} \right)' = -\frac{\cos(x)}{\sin^2(x)} = -\csc(x) \cdot \cot(x)$$

$$(\arcsin(x))' = \frac{1}{(\sin(t))'} = \frac{1}{\cos(t)} = \frac{1}{\sqrt{1-x^2}}$$

$$x = \sin(t)$$

$$(\arccos(x))' = -\frac{1}{\sqrt{1-x^2}}$$

$$(\arctan(x))' = \frac{1}{1+x^2} \quad , \quad (\operatorname{arccot}(x))' = -\frac{1}{1+x^2}$$



$$\boxed{(u(x))' = \frac{1}{x}} \quad , \quad (\log_a(x))' = \frac{1}{x \ln a}$$

$$\text{例: } y = e^{-x^2}, \quad y' = e^{-x^2} \cdot (-x^2)' = -2x e^{-x^2}$$

$$y = \sin\left(\frac{2x}{1+x^2}\right), \quad y' = \cos\left(\frac{2x}{1+x^2}\right) \cdot \left(\frac{2x}{1+x^2}\right)' = \cos\left(\frac{2x}{1+x^2}\right) \cdot \frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2}$$

$$= \cos\left(\frac{2x}{1+x^2}\right) \cdot \frac{2(1-x^2)}{(1+x^2)^2}$$

$$y = x^x = e^{x \ln x}$$

$$y' = e^{x \ln x} \cdot (x \ln x)' = e^{x \ln x} (1 + \ln x) = x^x (1 + \ln x)$$

双曲函数

$$(\operatorname{sh} x)' = \left(\frac{e^x - e^{-x}}{2}\right)' = \operatorname{ch}(x)$$

$$(\operatorname{ch} x)' = \left(\frac{e^x + e^{-x}}{2}\right)' = \operatorname{sh}(x)$$

$$\operatorname{th}(x)' = \left(\frac{\operatorname{sh}(x)}{\operatorname{ch}(x)}\right)' = \frac{1}{\operatorname{ch}^2(x)}$$

反双曲函数

$$(\operatorname{arsh} x)' = \left(\ln(x + \sqrt{1+x^2})\right)' = \frac{1}{\sqrt{1+x^2}}$$

$$(\operatorname{arch} x)' = \left(\ln(x + \sqrt{x^2-1})\right)' = \frac{1}{\sqrt{x^2-1}}, \quad x > 1$$

$$(\operatorname{arth} x)' = \left(\frac{1}{2} \ln \frac{1+x}{1-x}\right)' = \frac{1}{1-x^2}, \quad |x| < 1$$

2022.10.17

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§2.3 高阶导数

$$f(x), f'(x) = f^{(1)}(x), f''(x) = f^{(2)}(x), f'''(x) = f^{(3)}(x), \dots$$

求导 求导

① 定义, 记号.

$$f^{(n)}(x)$$

$$\frac{d^n f(x)}{dx^n}$$

$$\frac{d^n f(x)}{(dx)^n}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta^n f(x)}{(\Delta x)^n}$$

↑↑
Δⁿf(x) 高阶差商.

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$f''(x) = \lim_{\Delta x \rightarrow 0} \frac{f'(x+\Delta x) - f'(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{f(x+\Delta x+\Delta x) - f(x+\Delta x)}{\Delta x} - \frac{f(x+\Delta x) - f(x)}{\Delta x}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x+2\Delta x) - 2f(x+\Delta x) + f(x)}{(\Delta x)^2} \rightarrow \text{差商, } \Delta^2 f(x)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta^2 f(x)}{(\Delta x)^2}$$

不是 $f'(x)$ 的定义
但若 $f'(x)$ 存在, 则可以这么算.

$$\text{差商 } \Delta^n x_n, \Delta x_n = x_{n+1} - x_n$$

$$\Delta^2 x_n = \Delta(\Delta x_n) = (T-I)x_n$$

$$\Delta^2 x_n = (T-I)^2 x_n$$

$$= (T^2 - 2T + I)x_n$$

$$= x_{n+2} - 2x_{n+1} + x_n$$

$$\Delta^n x_n = (T-I)^n x_n$$

② [Leibniz Regel]

$$(u \pm v)^{(n)} = u^{(n)} \pm v^{(n)}$$

$$(uv)^{(n)} = \sum_{k=0}^n \boxed{\binom{n}{k}} u^{(n-k)} v^{(k)}, \quad \text{Leibniz's rule}$$

$\hookrightarrow C_n^k$

$$\left(\frac{1}{v}\right)^{(n)} = \dots$$

$$\left(\frac{u}{v}\right)^{(n)} = \dots$$

$[k \geq n]$

$$(uv)' = u'v + uv'$$

$$(uv)'' = ((uv)')'$$

$$= (u'v + uv')'$$

$$= u''v + u'v' + u'v' + uv''$$

$$= u''v + 2u'v' + uv''$$

③ $|f|$

$$x^k \xrightarrow{D} kx^{k-1} \xrightarrow{D} k(k-1)x^{k-2} \xrightarrow{D} k(k-1)(k-2)x^{k-3} \rightarrow \dots$$

$$(c)^{(n)} = 0$$

$$(x^k)^{(n)} = \begin{cases} 0, & n > k, \text{ } k \in \mathbb{Z}^+ \\ k(k-1)\dots(k-(n-1))x^{k-n}, & n \leq k, \text{ } k \notin \mathbb{Z}^+ \end{cases}$$

Leibniz's rule

$$(\sin(x))^{(n)} = \begin{cases} (-1)^{k+1} \cos(x), & n = 2k-1 \\ (-1)^k \sin(x), & n = 2k \end{cases}$$

$\sin(x + n \cdot \frac{\pi}{2})$

$$(\cos(x))^{(n)} = \cos(x + n \cdot \frac{\pi}{2}) = (\sin(x))^{(n+1)}$$

$$(e^x)^{(n)} = e^x$$

$$(a^x)^{(n)} = (\ln a)^n a^x$$

$$\begin{array}{ccc} \sin(x) & \xrightarrow{D} & \cos(x) \xrightarrow{D} -\sin(x) \\ & \uparrow & \downarrow D \\ & D & -(\cos(x)) \end{array}$$

$$\ln x \xrightarrow{D} \frac{1}{x} = x^{-1}$$

$$(\ln x)^{(n)} = \left(\frac{1}{x}\right)^{(n-1)}$$

$$(\ln(1+x))^{(n)} = \left(\frac{1}{1+x}\right)^{(n-1)} = (-1)^{n-1} \frac{(n-1)!}{(1+x)^n}$$

7/1

$$X_{n+2} = X_{n+1} + X_n \quad X_n = k_1 q^n + k_2 p^n$$

$$X_{n+2} - X_{n+1} - X_n = 0$$

$$y'' - y' - y = 0 \Rightarrow y = k_1 e^{\lambda x} + k_2 e^{-\mu x}$$

$$\textcircled{1} y = k_1 e^{\lambda x}, \quad y = k_2 e^{-\mu x}, \quad \lambda^2 - \lambda - 1 = 0$$

$$k_1 \lambda^2 e^{\lambda x} - k_1 \lambda e^{\lambda x} - k_1 = 0$$

$$k_1 e^{\lambda x} (\lambda^2 - \lambda - 1) = 0$$

$$y'' - 2y' + y = 0, \quad \lambda^2 - 2\lambda + 1 = 0, \quad \lambda = 1$$

$$\hookrightarrow y = k_1 e^x$$

$$y = k_2 x e^x$$

$$\Delta y = k_1 e^{\lambda_1 x} + k_2 e^{\lambda_2 x}$$

$$\begin{aligned} & \frac{e^{\lambda_1 x} - e^{\lambda_2 x}}{\lambda_1 - \lambda_2} \quad \lambda_1, \lambda_2 \rightarrow 1 \\ & \downarrow \\ & x e^{\lambda_2 x} \cdot \frac{e^{(\lambda_1 - \lambda_2)x} - 1}{(\lambda_1 - \lambda_2)x} \rightarrow \left(\frac{de^x}{dx} \right) \Big|_{t=0} \\ & \downarrow \\ & e^x \cdot x \end{aligned}$$

3.1.

$$\lim_{x \rightarrow +\infty} \frac{\log x}{\sqrt{x}} = 0$$

$$\lim_{t \rightarrow +\infty} \frac{t}{2^t} = 0$$

$$\lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$$

$\sqrt{x} = t$, 变成 $\frac{\ln t}{t} \rightarrow 0$ 的极限

$\ln t = y$, 变成 $\frac{y}{e^y} \rightarrow 0$

$$\frac{n}{(1+1)^n} = \frac{n}{1+n+\frac{n(n-1)}{2}+\dots+1} < \frac{n}{\frac{n(n-1)}{2}} = \frac{2}{n-1} < \varepsilon$$

讨论: ① $\lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0, \quad a > 1, \quad k \in \mathbb{R}$

$\Downarrow$

② $\lim_{n \rightarrow \infty} \frac{(\ln n)^m}{n^k} = 0, \quad k > 0, \quad m \in \mathbb{R}$

思路: 取 $a = 1+b, \quad b > 0$

$$a^n = (1+b)^n = 1 + nb + \dots + b^n$$

$$\geq \binom{n}{k+1} b^{k+1}$$

$$\frac{n^k}{a^n} \leq \frac{1}{b^{k+1}} \frac{n^k}{\underbrace{n(n-1)\dots(n-k)}} < \varepsilon$$

$\downarrow$
 $\sim n^{k+1}$

§ 2.4 隐函数求导

- 3个步骤

eg. $x = f(y)$
反函数不写出
是求表达式.

求导. 注意: $x = f(y(x))$

$$\downarrow$$

$$1 = f'(y) y'$$

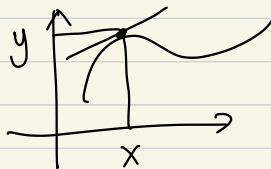
$$y' = \frac{1}{f'(y)}$$

参数方程

$$\begin{cases} x = x(t), & t \text{ 为参数.} \\ y = y(t) \end{cases}$$

$$t = t(x)$$

$$y = y(t(x)) = f(x)$$



$$f'(x) = \frac{y'(t)}{x'(t)}$$

$$\parallel$$

$$\frac{df(x)}{dx} = \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt} / \frac{dx}{dt}$$

显 $y = f(x)$

隐. $x = \sin(y), y = \arcsin(x)$

$$x = y + \sin(y) \Leftrightarrow y = g(x)$$

$$x = g(x) + \sin(g(x)) \quad \leftarrow \text{恒等式}$$

$$1 = g'(x) + \cos(g(x)) g'(x)$$

$$g'(x) = \frac{1}{1 + \cos(g(x))}$$

初等函数的反函数不一定是初等函数

① 写成恒等式

② 求导. 变形.

③ 高阶, 逐次求.

高阶导数.

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad \frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt}$$

$$\frac{d^2y}{dx^2} \neq \frac{d^2y}{dt^2} / \frac{d^2x}{dt^2}$$

特例:

$$\begin{cases} x = t \\ y = y(t) \end{cases} \quad \text{即 } y = y(x)$$

$$\begin{cases} x = x(t) \\ y = t \end{cases} \quad \text{即反函数.}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{y'(t)}{x'(t)} \right) = \frac{d}{dt} \left(\frac{y'(t)}{x'(t)} \right) / \frac{dx}{dt}$$

$$= \frac{y''(t)x'(t) - y'(t)x''(t)}{(x'(t))^3}$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d}{dt} \left(\frac{d^2y}{dx^2} \right) / \frac{dx}{dt} = \dots$$

例 9 计算由摆线 (图 2-9) 的参数方程

$$\begin{cases} x = a(t - \sin t), \\ y = a(1 - \cos t) \end{cases}$$

所确定的函数 $y = y(x)$ 的二阶导数.

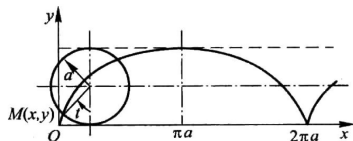
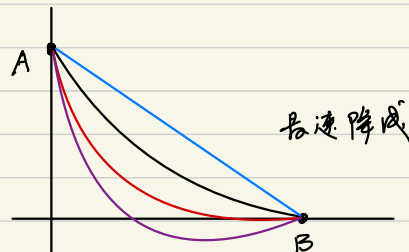
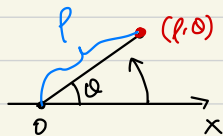


图 2-9



例: 极坐标. (ρ, θ)



$$\begin{cases} x = \rho(\theta) \cos(\theta) \\ y = \rho(\theta) \sin(\theta) \end{cases}$$

则 $\rho(0) = e^0$

心形线

$$\textcircled{1} \rho(\theta) = a(1 - \cos\theta)$$

$$\textcircled{2} \rho(\theta) = a(1 - \sin\theta) \quad \heartsuit$$

应用:

① 例6 求 $y = \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)}}$ 的导数.

$$\ln y(x) = \frac{1}{2} [\ln(x-1) + \ln(x-2) - \ln(x-3) - \ln(x-4)]$$

$$\frac{1}{y(x)} y'(x) = \frac{1}{2} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} \right]$$

$$y'(x) = \frac{y(x)}{2} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} \right]$$

$$= \frac{1}{2} \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)}} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} \right]$$

② $y = u(x)^{v(x)}, \quad y = e^{v(x) \ln u(x)}$

$$\ln y = v(x) \ln u(x)$$

$$\frac{y'}{y} = v'(x) \ln u(x) + \frac{v(x)}{u(x)} u'(x)$$

$$y' = u(x)^{v(x)} \left[v'(x) \ln u(x) + \frac{v(x)}{u(x)} u'(x) \right]$$

⑤ 相关变化率

2022.12.19
wed

§2.5 函数的微分

↑
增量与微分

微分

$$\Delta y = A \Delta x \quad (\text{直})$$

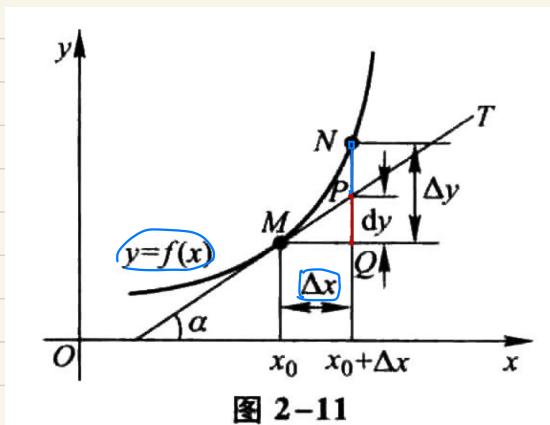
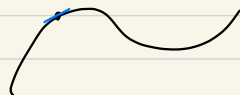
以直线为微分法

$$dy = A dx$$

① 定义. $\Delta y = f(x_0 + \Delta x) - f(x_0)$

若 $\Delta y = \boxed{A \Delta x} + o(\Delta x)$, 则称 $f(x)$ 在 x_0 处 可微

② $f(x)$ 在 x_0 处对 Δx 的微分, 记为 dy



关系: $\boxed{\text{可微} \Leftrightarrow \text{可导}} \Rightarrow \text{连续}$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{A \Delta x + o(\Delta x)}{\Delta x} = A + \lim_{\Delta x \rightarrow 0} \frac{o(\Delta x)}{\Delta x} = A$$

$$\boxed{dy = A(x) \Delta x = A dx}$$

$$y = f(x) = x$$

$$\frac{dy}{dx} = A$$

$$dx = dy = 1 \Delta x$$

② 微分的运算法则.

$$\begin{aligned} f(x), g(x), & \quad df(x) = f'(x) dx \\ f'(x), g'(x) & \quad dg(x) = g'(x) dx \end{aligned}$$

$$\begin{aligned} \underbrace{d(f \pm g)} &= df \pm dg = (f'(x) \pm g'(x)) dx \\ d(fg) &= (df)g + f(dg) = (fg)' dx = (f'g + fg') dx \\ d\left(\frac{f}{g}\right) &= \frac{(df)g - f(dg)}{g^2} = \frac{(f' dx)g + f(g' dx)}{g^2} \\ &= (df)g + f(dg) \\ \text{" } \left(\frac{f}{g}\right)' dx &= \frac{f'g - fg'}{g^2} dx \end{aligned}$$

$$d(cf) = c df$$

$$\begin{aligned} d(f(g)) &= f'(g) dg = \frac{df(g)}{dg} dg \\ \hookrightarrow (f \circ g)' dx &= (f'(g) g') dx \end{aligned}$$

$$\frac{df}{dx} = g, \quad df = g dx$$

$$\frac{d^2 f}{(dx)^2} = h \quad \text{有条件} \quad d^2 f = h dx^2$$

↑
有条件
可记不对.

- 211 微分形式不变性.

③ 高阶微分 $y = f(x)$

一阶 $dy = f'(x) dx$

二阶 $d^2y = d(dy) = d(f'(x) dx) = d(f'(x)) dx + f'(x) d(dx)$

$= f''(x) \underbrace{dx dx}_{(dx)^2} + f' \underbrace{d(dx)}_{d^2x}$ 是否等于 0. 取决于 dx 与 x 是否独立.

dx 与 x 独立
 $\quad \quad \quad = f''(x) (dx)^2$ $\rightarrow \frac{d(dx)}{dx} dx$

三阶, ..., n 阶. ...

例: $y = f(x) = \sin(x)$, $x = \varphi(t) = t^2$

求 d^2y

方法 2: $dy = f'(x) dx$

$d^2y = f''(x) (dx)^2 + f'(x) \cancel{d^2x}$
 $\quad \quad \quad = \cancel{\sin(x)} (dx)^2$ $d(dx) \neq 0$

$dx = \varphi'(t) dt = 2t dt$
 $\rightarrow d^2y = \cancel{4t^2 \sin(t^2)} (dt)^2$ dx 与 x 不独立.

方法 1: $y = \sin(x(t)) = \sin(t^2)$

$dy = \frac{d(\sin(t^2))}{dt} dt = \cos(t^2) 2t dt$

$d^2y = d(2t \cos(t^2) dt) = (2t \cos(t^2))' (dt)^2 + 2t \cos(t^2) \underline{\underline{=0}}$ t, dt 独立

$x = t^2$	$x \neq x_0$	$t = 0$
$dx = 2t dt$		$x = 0$
		$dx = 0$

④ 应用.

计算数学.

1. 误差计算

$$df(x) = g(x) dx$$

$$\Delta f(x) = f(x+\Delta x) - f(x) \approx g(x) \Delta x$$

$$f(x+\Delta x) \approx f(x) + f'(x) \Delta x$$

$$\sqrt{1.05} = \sqrt{1 + \underbrace{0.05}_{\Delta x}} \approx \sqrt{1} + \frac{1}{2\sqrt{1}} \Delta x = 1 + \frac{0.05}{2} = 1.025$$

$$\sqrt{1} = 1$$

2. 常用近似公式

$$(1+x)^\alpha \approx 1 + \alpha x$$

$$(1+x)^\alpha \sim 1 + \alpha x$$

$$\ln(1+x) \approx x.$$

误差.

绝对误差

相对误差

导数, 可导, 微商, 差商.

微分, 可微.

初等函数都是可导/微的. (自然定义域内)

eg.

$$= \begin{cases} e^{\ln x}, & x \neq 0 \\ 0, & x = 0 \end{cases} x^{-n} = e^{n \ln x}$$

$$|x| = \sqrt{x^2} = (x^2)^{\frac{1}{2}} = e^{\frac{1}{2} \ln x^2}, x \neq 0$$


自然定义域不包含 0.